



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

It's time to check previous projects!



Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features. Recognizes and implements the different types of architecture styles that can be included in the IT solution.

Can Do Statements:

- I can understand a written case of study related with Instrumentation software and architecture styles.
- I can answer questions related with the case of study.
- I can write a brief summary about the case study findings.

Ask your partner:

- What studies related to architecture styles do you know?
- What architecture style would you choose to conduct a study?





Let's Explore!

A professor talks about a case study!



Instructions: Read the text below.

In today's class, I'm going to present a case of study about instrumentation software and architecture styles. It was written by researchers from Tektronix, Inc. in a computer research laboratory.

The study intended to propose a way to develop a reusable system architecture for oscilloscopes, which companies needed to readjust due to changes in hardware capability. The problem was that many oscilloscopes were designed for specific purposes, which companies, later on, needed to modify. Therefore, if they wanted to reuse an item, they had to implement a general-purpose instrument. Another issue was that oscilloscopes weren't programmed within the instrument, so, they had to load a different software every time they needed to cause delays. To solve these problems, they needed to have a domain-specific software architecture.

The first question was, "what architecture style was more appropriate to accomplish what they wanted?" Well, I'm going to outline the stages, which researchers applied, that helped define the correct method. First, they tried an object-oriented model, which identified waveforms, to clarify data types. However, two more questions appeared, "Are measurements associated or represented externally with the type of data?", and "What objects does the interface user have to talk about?". Since all the functions could be defined into groups, these questions were answered by using a layered model. Unfortunately, layers often withstood interaction among functions. Finally, the pipe and filter model suited the expectations after further specialization and development of the style.

If we wanted to summarize this case study, we would say that it took many attempts and deep research to find an appropriate model. One of the main findings was that pipes and filters can be structured for specific needs. Also, any architecture style has to be taken "pure" and then it can be offset depending on the purpose.





Let's Practice!

What is the professor doing with the case of study?

Instructions: Select the correct answer to complete the sentences.

1. Has the professor, _____ is working in the instrumentation software project, presented the case of study?
 - a. which
 - b. who
 - c. that
2. Does this case of study, _____ is shown by the professor, have reusable purpose?
 - a. who
 - b. which
 - c. where
3. What is the problem _____ the case of study wants to show?
 - a. that
 - b. where
 - c. who
4. Does the laboratory, _____ the professor works, has to change its hardware capability?
 - a. which
 - b. who
 - c. where
5. What was the _____ the case of study has shown? _____ the
 - a. solution that
 - b. solution where
 - c. solution who



Let's Practice!

Case of Studies have many important words to remember.

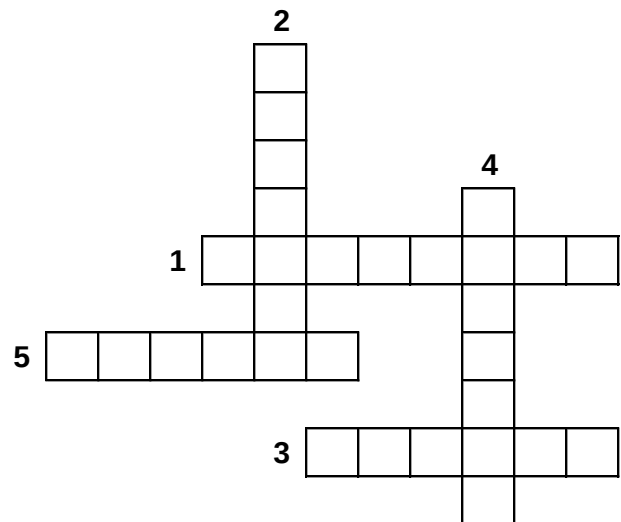
Instructions: Read the clues. Complete the crossword.

Down

2. First, if I wanted to reuse an item, I would have to implement a general-_____ instrument.
4. I would say that this question was answered by using a _____ model since all the functions could be defined into groups.

Across

1. I will probably choose everything about instrumentation _____ and architecture styles.
3. I would try an _____-oriented model, which identified waveforms, to clarify data types.
5. I would say that the pipe and _____ model is suited the expectations after further development of the style.





Extension Activity

Let's analyze the next problem.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about any case of study related to instrumentation software and architecture style.
- Take notes about the case of study you are reading
- Write a brief report about the case of study that you have read.
- Share your report with your partners.

Self-Assessment

Yes

No

- I search information about case of study related to instrumentation software and architecture style.

☐ ☐

- I write a brief report about the case of study that I have chosen.

☐ ☐

- I share y report with my partners.

☐ ☐


Target Words

- **propose** (verb): to offer ideas in order to accomplish something
e.g. The study intended to propose a way to develop a reusable system architecture for oscilloscopes, which companies needed to readjust due to changes in hardware capability.
- **summarize** (verb): to present a topic briefly
e.g. If we wanted to summarize this case study, we would say that it took many attempts and deep research to find an appropriate model.
- **withstand** (verb): to resist to something
e.g. Unfortunately layers often withstood interaction among functions.
- **offset** (verb): to compensate in order to balance something
e.g. Any architecture style has to be taken "pure" and then it can be offset depending on the purpose.
- **outline** (noun): to give the main features of something
e.g. I'm going to outline the stages, which researchers applied, that helped define the correct method.
- **oscilloscope** (noun): Instrument that allows observation of signal voltages
e.g. Another issue was that oscilloscopes weren't programmed within the instrument, so, they had to load a different software every time they needed to cause delays.



Learning Points

Object Relative Clauses

You use **relative clauses** to provide extra information about an **object**.

In order to create a sentence, we need to use relative pronouns that can replace an **object**. If a noun or a pronoun is after the relative clause, then it is an **object relative clause**.

These are the relative pronouns that you can use:

Object relative pronouns:

Who / Which / Where/ That

Defining Relative Clauses:

We add important information that is relevant and is that is needed to understand who or what we are referring to.

Examples:

*Case studies **which** experts conducted showed great benefits.*

*The architecture style **that** was chosen didn't work.*

No-defining Relative Clauses:

In this kind of clauses **we don't use** the relative pronoun **that**. The extra information that you want to provide is irrelevant and you have to put it between two commas. This means that the sentence keeps the message with or without the non-defining relative clause.

Look at this sentence:

The case studies had very interesting hypothesis.

Let's add a **non-defining relative clause**:

*The case studies, **which** I read, had very interesting hypothesis.*

Another example is:

*The architecture style, **which** I thought would be the best option, ended up being the last option for the case study.*

Examples

- *The problem was that many oscilloscopes were designed for specific purposes, which companies, later on, needed to modify.*
- *I'm going to outline the stages, which researchers applied, that helped define the correct method.*
- *They tried an object-oriented model, which identified waveforms, to clarify data types.*





Learning Points

What is a case of study?

A **case of study** is a record of a real or hypothetical situation used to describe the influences of decisions.

Case studies are implemented to describe a behavior of an individual or a group. They are considered a formal research method in many disciplines. The word “case” refers to the situation that is being studied.

There are different types of case studies which could be divided into two main categories; **historical** and **problem orientated**. They could be **historical** where the causes and the consequences of a situation are analyzed so, you are outside the situation. On the other hand, if you want to analyze a current situation and want to find out answers, the case study would be **problem orientated** as you are part of process.

Some subdivisions of case studies help us identify what we want to accomplish. As an investigator you may want to choose one of the following categories:

Illustrative: These case studies are descriptive. They intend to contextualize readers about an unknown topic.

Exploratory: A case study designed as the first stage before a big investigation. It helps finding questions and possible conclusions.

Cumulative: These studies collect information from other studies in order to obtain generalized concepts.

Critical Instance: Case studies are implemented to analyze cause and effect questions under unique situations.

Examples

- *I'm going to present a case of study about instrumentation software and architecture styles. It was written by researchers from Tektronix, Inc. And a computer research laboratory.*
- *The study intended to propose a way to develop a reusable system architecture for oscilloscopes which companies needed to readjust due to changes in hardware capability.*

